

RB002 Week 5 — English Worksheet

Topic: Quest Items — Pick Up Crystals · **Course:** Story Quest · **Time:** about 45 minutes

This week your player picks up **quest items**. A crystal counts up when a player touches it, then disappears with `Destroy()`. A **debounce** stops double-counting, and a **TextLabel** on screen shows the count.

1 · Predict 🧠

Read each script. Before you imagine it running in Roblox Studio, write what you think will happen.

```
local crystal = script.Parent
local Players = game:GetService("Players")

crystal.Touched:Connect(function(hit)
    local player = Players:GetPlayerFromCharacter(hit.Parent)
    if player then
        player.leaderstats.Crystals.Value += 1
        crystal:Destroy()
    end
end)
```

A player walks into the crystal. What happens to the Crystals counter? What happens to the crystal part?


```
local count = 3
print("Crystals: " .. count)
```

What exactly prints in the Output window? Write it letter by letter.


```
local touched = false

crystal.Touched:Connect(function(hit)
    local player = Players:GetPlayerFromCharacter(hit.Parent)
    if player and not touched then
        touched = true
        print("Picked up!")
    end
end)
```

The player's foot AND leg both touch the crystal at almost the same time. How many times does "Picked up!" print? Why?

2 · Spot the Bug 🐛

Each script below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — Touching the crystal should add **exactly 1** to the counter. Right now it sometimes adds 2 or 3, because `Touched` fires many times before `Destroy()` finishes.

```
crystal.Touched:Connect(function(hit)
    local player = Players:GetPlayerFromCharacter(hit.Parent)
    if player then
        player.leaderstats.Crystals.Value += 1
        crystal:Destroy()
    end
end)
```

Hint: add a **debounce** variable that starts `false`.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The label should show `Crystals: 4`. Right now the script turns red and shows an error.

```
label.Text = "Crystals: " + crystals.Value
```

Hint: in Luau, `+` is only for math. Joining text needs a different symbol.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Only **players** should collect the crystal. Right now the crystal destroys itself on ANY touch — even a rolling ball — and then the script errors because `player` is `nil`.

```
crystal.Touched:Connect(function(hit)
    crystal:Destroy()
    local player = Players:GetPlayerFromCharacter(hit.Parent)
    player.leaderstats.Crystals.Value += 1
end)
```

Hint: check `if player then` **before** you count or destroy anything.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Now open your Story Quest game in Roblox Studio. Build at least 3 crystals that players can pick up, with a counter and a TextLabel that shows the count. When you finish, come back here.

Send a photo or video of your crystals and your label, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

First, I ...

My crystal script uses **Touched** when ...

The counter went up when ...

My label shows ...

One tricky moment was when ...

To fix it, I ...

4 · Record Your Walkthrough 🎥

Now take a video on your phone (or a parent's phone) while you play your game in Studio. Talk through it like you are teaching someone who has never seen it. Try to use these words: **Touched**, **debounce**, **counter**, **Destroy**, **label**.

1. Show your crystals before you touch them, then press Play.
2. Pick up one crystal and show the counter going up by exactly 1.
3. Point at your label and read what it says out loud.
4. Open your script and read the debounce lines. Say why they stop double-counting.
5. Say in your own words what `Destroy()` does.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.